

Construction and Design of MPAC Codes

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Abstract

This paper proposes modified polarization-adjusted convolutional (MPAC) codes and their hybrid decoding that achieves an improved performance-complexity tradeoff. For MPAC codes, only a subset of the information bits undergo the convolutional transform. The output is then combined with the remaining information bits for the inner polar transform. Correspondingly, the convolutionally transformed bits are recovered by Fano decoding, while the remaining information bits are recovered by the successive cancellation (SC) decoding, constituting the hybrid Fano-successive cancellation (HFSC) decoding. The MPAC codes are further designed by the coset-wise analysis that characterizes the number of minimum weight codewords (MWCs). It is discovered that a partially convolutional transform can improve the codeword through utilizing the row combinations of the frozen set efficiently. This property enables the MPAC codes to outperform their prototype polarization-adjusted convolutional (PAC) codes and cyclic redundancy check (CRC)-polar codes. Furthermore, MPAC codes can be optimized by reducing the number of MWCs. Our numerical results demonstrate that, with a similar decoding complexity budget, the MPAC codes offer competent decoding performance when compared with PAC codes using Fano decoding and CRC-polar codes using SC list (SCL) decoding.

Index Terms

Fano decoding, minimum weight distribution, polar codes, polarization-adjusted convolutional codes.

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I. INTRODUCTION

Polar codes have been proven to achieve capacity of the binary input discrete memoryless channel (BI-DMC), under the prerequisites of approaching infinite codeword length and the successive cancellation (SC) decoding [1]. However, for short-to-medium length polar codes, the SC decoding performances remain limited. A notable improvement can be achieved through the SC list (SCL) decoding [2] [3], which keeps L distinct instances of the SC decoding paths. The SCL decoding can approach the maximum likelihood (ML) decoding performance with a sufficiently large list size L . Further improvement can be obtained by concatenating polar codes with the cyclic redundancy check (CRC) codes [2] [3], the parity check (PC) codes [4] [5] or the convolutional codes [6] [7].

The polarization-adjusted convolutional (PAC) codes [7] concatenate an outer rate-1 convolutional code with an inner polar code. It is shown that the normal approximation (NA) bound can be approached by PAC codes using Fano decoding [7]. However, PAC codes are usually constructed through the Reed-Muller (RM) rate profiling [7], which yields limited code rates. Addressing this, the existing polar code rate profiling schemes [1] [8]–[11] can be utilized for selecting the reliable subchannels, while the RM-polar rate profiling [12] [13] can yield both a flexible rate choice and improved codeword weight distribution. By further considering the effect of convolutional transform, performances of PAC codes can be improved. By considering subchannel utilizations, Liu *et al.* [14] proposed the weighted sum (WS) metric for a more flexible rate profiling. It generalizes the RM rate profiling to design PAC code for any desired rate. Several rate profiling designs have also been proposed in order to optimize the decoding error probability [15] [16], reducing the Fano decoding complexity [17] [18]. The minimum weight distribution (MWD) contains knowledge of the code's minimum Hamming distance and the number of its minimum weight codewords (MWCs). Moreover, by considering the transpose of the convolutional transform matrix [19] and formation of MWCs [20], the MWD of PAC codes can also be improved. It leads to an enhanced ML decoding performance for the codes.

Fano decoding [21] is an efficient algorithm in terms of its required hardware implementation resources, including memory and computation resources. However, when the received information is unreliable, Fano decoding may suffer from a high decoding latency and complexity. To reduce the Fano decoding complexity, Moradi *et al.* [22] proposed the path metric function that is parameterized by the subchannel cutoff rates. Meanwhile, Rowshan *et al.* [13] proposed several

tree search strategies to reduce the decoding complexity. They both enable the Fano decoding to achieve a better performance-complexity tradeoff. To further curb worst-case decoding latency and complexity, Fano decoding can be performed with a computational threshold [22]–[24]. But it inevitably leads to a decoding performance degradation. So far, limited efforts have been deployed through redesigning PAC codes in achieving a better performance-complexity tradeoff.

It is known that a code's ML decoding performance is determined by its weight distribution. Both polar codes and PAC codes can be designed through optimizing their weight distributions [19], [20], [25]–[27]. With an exhaustive search, weight distribution of both polar codes and PAC codes can be computed, but only up to the length of 64 bits [25] [27]. Utilizing the recursive structure of polar codes, the probabilistic methods [28] [29] can estimate the codes' weight distribution. With a similar approach, Li *et al.* [30] analyzed the average weight distribution of the pre-transformed polar codes which include PAC codes. Overall, it remains challenging to compute a code's entire weight distribution. Consequently, the MWD becomes a predominant metric in asymptotic performance analysis, as it dictates the error probability floor at high signal-to-noise ratio (SNR) regimes [31]. For a family of polar codes that are defined by the partial order property, Bardet *et al.* [32] provided an explicit formula for computing the number of MWCs. For the same family of codes, Rowshan *et al.* [20] characterized the row combinations of the MWCs under the coset-wise framework using a set-theoretic approach. It can be further generalized for analyzing the MWD of PAC codes [33]. Besides the above mentioned theoretical analysis, the MWD of both polar codes and PAC codes can also be numerically estimated. Li *et al.* [34] proposed the SCL-based enumeration method to evaluate the number of MWCs. But it requires a huge memory and complexity since the decoding output list size L needs to be sufficiently large.

This paper proposes the modified PAC (MPAC) codes and their hybrid Fano-SC (HFSC) decoding. This coding scheme was first proposed in our previous work of [35], which performs convolutional transform only for a portion of information bits. It limits the number of nodes that are allowed for backtracking, yielding a better performance-complexity tradeoff than the relevant coding schemes. In this work, the MPAC code design is further proposed based on the coset-wise analysis of its MWD, leading to the approximated union bound (AUB)-optimal MPAC codes that have a better MWD than PAC codes and CRC-polar codes. It also shows the effectiveness of the partially convolutional transform design for improving the MWD. Major contributions of this work are summarized as follows:

- The MPAC codes are proposed. In the coding scheme, only a subset of information bits undergo the convolutional transform. Its output bits are then concatenated with the remaining information bits for the inner polar transform.
- The HFSC decoding is further proposed to decode information bits. It recovers the convolutionally transformed information bits and the remaining information bits through Fano decoding and SC decoding, respectively. Our numerical results show that with a budgeted complexity, the proposed scheme yields a better performance-complexity tradeoff than Fano decoding of PAC codes and SCL decoding of CRC-polar codes.
- The coset-wise analysis of the MWD of MPAC codes is presented, which leads to design the AUB-optimal MPAC codes. By considering the impact of frozen bits in MPAC codes, it characterizes the row combinations of the MWCs. By further validating generation of the characterized codewords, the exact volume of MWCs in MPAC codes can be obtained. For a wide range of code parameters, the designed MPAC codes yield a better MWD than PAC codes and CRC-polar codes.

The rest of this paper is organized as follows. Section II provides the background knowledge on polar codes and PAC codes. Section III proposes the MPAC codes and their HFSC decoding. Section IV analyzes the MWD of MPAC codes in the coset-wise framework. Section V presents the design of AUB-optimal MPAC codes and their numerical results. Finally, Section VI concludes this paper. Ahead of the rest sections, we first state our notation fashion as follows.

Let $\mathbb{F}_2 = \{0, 1\}$ and $\oplus$ denote the binary field and its addition operator, respectively. We also let $\oplus$ denote the addition operator between the binary vectors. Let $[l, u]$ denote the set $\{l, l+1, \dots, u\}$, where $l < u$. For a set $\mathcal{A} \subseteq [0, N-1]$, its cardinality and complementary set are denoted by $|\mathcal{A}|$ and $\mathcal{A}^c$, respectively. Further let $\underline{a}_i^j$, where $i < j$, denote the vector $(a_i, a_{i+1}, \dots, a_j)$. Given a vector $\underline{a}_0^{N-1}$ and a set $\mathcal{A} \subseteq [0, N-1]$, we use $\underline{a}_{\mathcal{A}}$ to denote its subvector $(a_i \mid i \in \mathcal{A})$. Note when $\mathcal{A} = [0, N-1]$, $\underline{a}_0^{N-1} = \underline{a}_{\mathcal{A}}$. The binary representation of an integer $i \in [0, 2^n - 1]$ is defined as $\text{bin}(i) = i_{n-1} \dots i_1 i_0$, where i_{n-1} is the most significant bit and $i = \sum_{b=0}^{n-1} i_b 2^b$. The support of a vector $\underline{a}_0^{N-1}$ is the index set of its non-zero coordinates, i.e., $\text{supp}(\underline{a}_0^{N-1}) \triangleq \{i \in [0, N-1] \mid a_i \neq 0\}$. Let $\mathcal{S}_i$ denote the support of $\text{bin}(i)$, i.e., $\mathcal{S}_i \triangleq \{b \in [0, n-1] \mid i_b \neq 0\}$. Furthermore, let $\mathcal{S}_j \setminus \mathcal{S}_i$ denote all elements in the support of $\text{bin}(j)$ but are not in the support of $\text{bin}(i)$. The Hamming weight of a vector $\underline{c}_0^{N-1} \in \mathbb{F}_2^N$ is denoted as $w(\underline{c}_0^{N-1}) \triangleq |\text{supp}(\underline{c}_0^{N-1})|$. Given two vectors $\underline{c} = (c_0, c_1, \dots, c_{N-1}) \in \mathbb{F}_2^N$ and $\underline{c}' = (c'_0, c'_1, \dots, c'_{N-1}) \in \mathbb{F}_2^N$, the Hamming distance between them is defined as $d(\underline{c}, \underline{c}') = |\{i \in [0, N-1] \mid c_i \neq c'_i\}|$.

II. POLAR CODES AND PAC CODES

This section provides the prerequisites for this work, including polar codes and its SC decoding, PAC codes and its Fano decoding.

A. Approximated Union Bound (AUB)

A K -dimensional subspace $\mathcal{C}$ of $\mathbb{F}_2^N$ is called an $(N, K, d_{\min})$ binary linear block code, where N and K are the length and dimension of the code, respectively, and the minimum distance $d_{\min}$ of the code is defined as

$$d_{\min} = \min_{\underline{c}, \underline{c}' \in \mathcal{C}, \underline{c} \neq \underline{c}'} d(\underline{c}, \underline{c}').$$

For a binary input additive white Gaussian noise (BI-AWGN) channel, union bound [31] on the ML decoding frame error rate (FER), denoted as P_e , of a binary linear block code is

$$P_e \leq \sum_{d=d_{\min}}^N A_d Q\left(\sqrt{2d \cdot R \cdot E_b/N_0}\right), \quad (1)$$

where A_d denotes the number of weight- d codewords, $Q(y) = \frac{1}{\sqrt{2\pi}} \int_y^\infty e^{-\frac{\vartheta^2}{2}} d\vartheta$ is the tail distribution function of the standard normal distribution, $R = K/N$ is the code rate and E_b/N_0 is the SNR per information bit. The MWD is defined by both $d_{\min}$ and $A_{d_{\min}}$. As SNR increases, MWD becomes dominant for (1). Hence, the union bound can be approximated by

$$P_e \approx A_{d_{\min}} Q\left(\sqrt{2d_{\min} \cdot R \cdot E_b/N_0}\right). \quad (2)$$

The above AUB is often utilized to assess the operational performance of a designed binary linear block code.

B. Polar Codes and SC Decoding

Polar codes are founded on channel capacity polarization which is the consequence of channel combining and splitting [1]. Given a BI-DMC $W : \mathcal{X} \rightarrow \mathcal{Y}$, with $\mathcal{X} \in \mathbb{F}_2$ and $\mathcal{Y} \in \mathbb{R}$, a set of N polarized subchannels $W_N^{(i)} : \mathcal{X} \rightarrow \mathcal{Y}^N \times \mathcal{X}^{i-1}$ can be obtained through the channel polarization process, where $N = 2^n$, $n \in \mathbb{N}^+$ and $i \in [0, N-1]$. Let $I(W)$ and $I(W_N^{(i)})$ denote the symmetric capacity of channel W and subchannel $W_N^{(i)}$, respectively. Channel polarization results in most of the subchannels $W_N^{(i)}$ becoming either noiseless with $I(W_N^{(i)}) \rightarrow 1$, or heavily noisy with $I(W_N^{(i)}) \rightarrow 0$.

An (N, K) polar code is constructed by the rate profiling and polar transform $\mathbf{G}_p = \mathbf{F}^{\otimes n}$, where $\mathbf{F} = ((1, 0), (1, 1))^T \in \mathbb{F}_2^{2 \times 2}$ is the kernel matrix and $\otimes$ denotes the Kronecker product. Rate profiling with parameters $(N, K, \mathcal{I})$ embeds the information vector $\underline{m}_0^{K-1} \in \mathbb{F}_2^K$ into a carrier vector $\underline{u}_0^{N-1} \in \mathbb{F}_2^N$. It is defined by the information set $\mathcal{I} \subseteq [0, N-1]$ and the frozen set $\mathcal{I}^c = [0, N-1] \setminus \mathcal{I}$, such that $\underline{u}_{\mathcal{I}} = \underline{m}_0^{K-1}$ and $\underline{u}_{\mathcal{I}^c} = \underline{0}$. After rate profiling, the codeword $\underline{c}_0^{N-1} \in \mathbb{F}_2^N$ can be generated by polar transform as $\underline{c}_0^{N-1} = \underline{u}_0^{N-1} \mathbf{G}_p$. By denoting the polar transform matrix $\mathbf{G}_p$ as $\mathbf{G}_p = [\mathbf{g}_0^T, \mathbf{g}_1^T, \dots, \mathbf{g}_{N-1}^T]^T$, the polar codeword can be defined as

$$\underline{u}_0^{N-1} \mathbf{G}_p = u_0 \mathbf{g}_0 \oplus u_1 \mathbf{g}_1 \oplus \dots \oplus u_{N-1} \mathbf{g}_{N-1}. \quad (3)$$

With rate profiling that sets $\underline{u}_{\mathcal{I}^c} = \underline{0}$, $\{\mathbf{g}_i \mid i \in \mathcal{I}\}$ forms a basis of the polar codebook, denoted as $\mathcal{C}(\mathcal{I})$.

Polar codes are decoded by the SC algorithm in which the information bits are estimated successively based on their decision log-likelihood ratios (LLRs). Let us assume that codeword $\underline{c}_0^{N-1}$ is transmitted through a noisy channel using binary phase shift keying (BPSK) modulation. Let $\underline{y}_0^{N-1} = (y_0, y_1, \dots, y_{N-1}) \in \mathbb{R}^N$ and $\hat{\underline{u}}_0^{N-1} = (\hat{u}_0, \hat{u}_1, \dots, \hat{u}_{N-1}) \in \mathbb{F}_2^N$ denote the received symbol vector and the estimated carrier vector, respectively. Over the SC decoding trellis, the decision LLR of the estimated bit $\hat{u}_i$ is defined as

$$L_N^{(i)} = \ln \frac{P(\underline{y}_0^{N-1}, \hat{\underline{u}}_0^{i-1} \mid \hat{u}_i = 0)}{P(\underline{y}_0^{N-1}, \hat{\underline{u}}_0^{i-1} \mid \hat{u}_i = 1)}, \quad (4)$$

where $P(\underline{y}_0^{N-1}, \hat{\underline{u}}_0^{i-1} \mid \hat{u}_i)$ is the transition probability of $W_N^{(i)}$.

Estimation $\hat{u}_i$ can be successively made based on the decision LLR of (4). For $i \in \mathcal{I}^c$, $\hat{u}_i = 0$ as they are frozen. For $i \in \mathcal{I}$, u_i is an information bit which can be estimated based on a bitwise ML decision rule as

$$\hat{u}_i = \Upsilon(L_N^{(i)}) = \begin{cases} 0, & \text{if } L_N^{(i)} \geq 0; \\ 1, & \text{otherwise.} \end{cases} \quad (5)$$

The SC decoding complexity is measured as the number of LLR computations, i.e., $N \log_2 N$.

C. PAC Codes and Fano Decoding

A PAC code concatenates an outer rate-1 convolutional transform with an inner polar transform, as shown in Fig. 1.

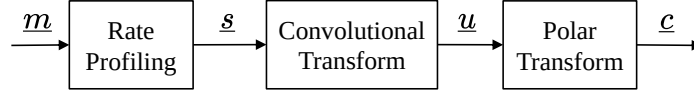


Fig. 1. Block diagram of PAC codes.

Algorithm 1: Generation of a single convolutional output bit, `conv1b(·)`

Input: v, S, g_0^t ;
Output: u, S ;
1 Initialize $u = v \cdot g_0$;
2 For $j = 1 \rightarrow |S|$ **do** // $|S|$ previous inputs
3 **If** $g_j = 1$ **then**
4 $u = u \oplus S[j]$;
5 $S = (v, S[1], \dots, S[t-1])$; // update shift register state
6 Return $[u, S]$;

This concatenation protects the message bits in the inner code through utilizing their correlation provided by the convolutional transform, yielding an improved decoding performance.

Let $\underline{s}_0^{N-1} \in \mathbb{F}_2^N$ and $\underline{u}_0^{N-1} \in \mathbb{F}_2^N$ denote the input and output vectors of the convolutional transform, respectively. To encode an (N, K) PAC code, information vector $\underline{m}_1^K$ is first embedded into $\underline{s}_0^{N-1}$ under the rate profiling with parameters $(N, K, \mathcal{I})$, such that $\underline{s}_{\mathcal{I}} = \underline{m}_0^{K-1}$ and $\underline{s}_{\mathcal{I}^c} = \underline{0}$. $\underline{u}_0^{N-1}$ is further generated by the convolutional transform as

$$u_i = \sum_{j=0}^t g_j s_{i-j}, \quad (6)$$

where $\underline{g}_0^t = (g_0, g_1, \dots, g_t) \in \mathbb{F}_2^{t+1}$ are coefficients of the convolutional generator polynomial. Let $\underline{S} = (S[1], S[2], \dots, S[t])$ denote state of the convolutional shift registers. Algorithm 1 (`conv1b(·)`) describes the generation of a convolutional output bit u based on the current state S and the input bit v . Equivalently, the convolutional transform can be represented in an upper-triangular Toeplitz matrix $\mathbf{G}_c \in \mathbb{F}_2^{N \times N}$, whose rows are formed by shifting the vector $\underline{g}_0^t$. Hence, $\underline{u}_0^{N-1} = \underline{s}_0^{N-1} \mathbf{G}_c$. Finally, the PAC codeword $\underline{c}_0^{N-1}$ is further generated by $\underline{c}_0^{N-1} = \underline{u}_0^{N-1} \mathbf{G}_p$. Therefore, convolutional coded bits $\underline{u}_0^{N-1}$ are transmitted through the polarized subchannels.

Fano decoding was first adopted to decode PAC codes, in which the constraints of the convolutional transform is imposed [7].

It can be seen as a depth-first search algorithm with the path metrics serving as the reliability

indicators.

In this paper, we use the optimal path metric function [22], which can be computed layer-by-layer as

$$M(\hat{\underline{u}}_0^i) = M(\hat{\underline{u}}_0^{i-1}) + 1.0 + \log_2 P(\hat{u}_i | \underline{y}_0^{N-1}, \hat{\underline{u}}_0^{i-1}) - E_0(1, W_N^{(i)}), \quad (7)$$

where $P(\hat{u}_i | \underline{y}_0^{N-1}, \hat{\underline{u}}_0^{i-1})$ and $E_0(1, W_N^{(i)})$ are the *a posteriori* probability of $\hat{u}_i$ and the cutoff rate of subchannel $W_N^{(i)}$, respectively.

For the BI-AWGN channel, $E_0(1, W_N^{(i)})$ can be obtained by [36]

$$E_0(1, W_N^{(i)}) = 1 - \log_2(1 + \exp(-1/(2(\sigma_N^{(i)})^2))), \quad (8)$$

where $(\sigma_N^{(i)})^2$ is the simulated noise variance of $W_N^{(i)}$. It can be estimated via the Gaussian approximation (GA) [10]. By adjusting threshold T dynamically with a step size Δ [21], Fano decoding compares $M(\hat{\underline{u}}_0^i)$ in controlling the decoding moves as:

- 1) If $\{M(\hat{\underline{u}}_0^i)\}_{\max} \geq T$, move forward to path $\{\hat{\underline{u}}_0^i\}_{\max}$;
- 2) If $\{M(\hat{\underline{u}}_0^i)\}_{\max} < T$, move backward to the root node of the tree, or the ancestor node $\hat{u}_{i'}$ which satisfies $\{M(\hat{\underline{u}}_0^{i'})\}_{\min} > T$, where $i' \leq i$, and the path $\{\hat{\underline{u}}_0^{i'}\}_{\min}$ has not been visited.

At the beginning of the decoding, state of the shift registers and the path metric are initialized as $\underline{S} = \underline{0}$ and $M(\hat{\underline{u}}_0^{-1}) = 0$, respectively. Assuming that bit $\hat{u}_i$ ($i \in \mathcal{I}$) has been explored by Fano decoding and the current state of shift registers is $\underline{S}$, the information bit $\hat{s}_i$ can be determined by the convolutional constraint as

$$\hat{s}_i = \lambda(\hat{u}_i, \underline{S}) = \begin{cases} 1, & \text{if } \text{conv1b}(1, \underline{S}, \underline{g}_0^t) = [\hat{u}_i, \underline{S}']; \\ 0, & \text{otherwise,} \end{cases} \quad (9)$$

where $\underline{S}'$ means the next shift registers corresponding to the output estimation $\hat{u}_i$. Furthermore, the next state $\underline{S}$ is updated by $\underline{S}'$.

Once the decoding reaches a leaf of the tree, the complete estimations of $\hat{\underline{u}}_0^{N-1}$ and $\hat{\underline{s}}_0^{N-1}$ are obtained.

III. MPAC CODES AND HFSC DECODING

This section introduces the MPAC codes and their HFSC decoding, which can yield a good performance-complexity tradeoff. Construction of the MPAC encoding includes two rate profiles which are defined by the proposed improved RM-polar (iRMP) orderings and the Gaussian

approximation (GA), respectively. The HFSC decoding integrates Fano decoding and SC decoding to recover the information bits.

A. Construction of MPAC Codes

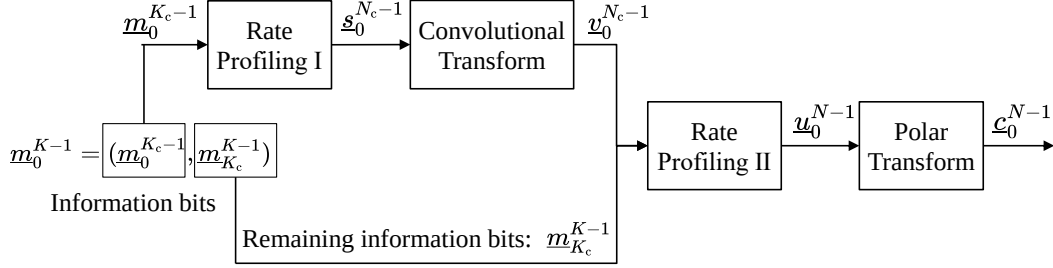


Fig. 2. Block diagram of the $(N, K)-(N_c, K_c)$ MPAC codes.

For an (N, K) MPAC code, it needs to be further specified by the *convolutional parameters* (N_c, K_c) , where N_c is the dimension of the convolutional transform as $\mathbf{G}'_c \in \mathbb{F}_2^{N_c \times N_c}$ and K_c is the number of information bits that undergo the transform. Fig. 2 shows the encoding block diagram of an $(N, K)-(N_c, K_c)$ MPAC code. The convolutional parameters should satisfy

$$K - K_c + N_c \leq N. \quad (10)$$

The equality holds when the MPAC code becomes the (N, K) PAC code, i.e., when $K_c = K$ and $N_c = N$. Hence, the MPAC codes can be considered as a more general case of PAC codes where the convolutional transform is performed on a portion of the information bits.

To encode an $(N, K)-(N_c, K_c)$ MPAC code, the information bits $\underline{m}_0^{K-1}$ is partitioned into $\underline{m}_0^{K-1} = (\underline{m}_0^{Kc-1}, \underline{m}_{Kc}^{K-1})$, where $\underline{m}_0^{Kc-1}$ and $\underline{m}_{Kc}^{K-1}$ denote the information bits that will undergo the convolutional transform and the remaining information bits, respectively. Furthermore, rate profiling I and II are utilized to construct the input vectors of the convolutional transform and the polar transform, denoted as $\underline{s}_0^{Nc-1}$ and $\underline{u}_0^{N-1}$, respectively. The information bits $\underline{m}_0^{Kc-1}$ will first be embedded into the vector $\underline{s}_0^{Nc-1}$ under the rate profiling I with parameter $(N_c, K_c, \mathcal{D})$, where $\mathcal{D} \subseteq [0, N_c]$ and $|\mathcal{D}| = K_c$. $\underline{s}_0^{Nc-1}$ is constructed as $\underline{s}_0^{Nc-1} = (\underline{s}_{\mathcal{D}}, \underline{s}_{\mathcal{D}^c})$, where $\underline{s}_{\mathcal{D}} = \underline{m}_0^{Kc-1}$ and $\underline{s}_{\mathcal{D}^c} = \underline{0}$.

The convolutional codeword is further generated by $\underline{v}_0^{Nc-1} = \underline{s}_0^{Nc-1} \mathbf{G}'_c$. It will be concatenated with the remaining information bits $\underline{m}_{Kc}^{K-1}$ to be transmitted through the polarized subchannels.

The subchannels are chosen for transmitting information bits based on their reliabilities. Let $\mathcal{A} \subseteq [0, N - 1]$ denote the index set of the $K - K_c + N_c$ most reliable subchannels. Among set $\mathcal{A}$, the N_c least reliable subchannels are indexed by set $\mathcal{P}$.

Consequently, under rate profiling II, the vector $\underline{u}_0^{N-1}$ is partitioned into

$$\underline{u}_0^{N-1} = (\underline{u}_{\mathcal{A} \setminus \mathcal{P}}, \underline{u}_{\mathcal{P}}, \underline{u}_{\mathcal{A}^c}), \quad (11)$$

where $\underline{u}_{\mathcal{A} \setminus \mathcal{P}} = \underline{m}_{K_c}^{K-1}$, $\underline{u}_{\mathcal{P}} = \underline{v}_0^{N_c-1}$ and $\underline{u}_{\mathcal{A}^c} = \underline{0}$ are the remaining information bits, the convolutional codeword and the frozen bits, respectively. Finally, the MPAC codeword is generated by $\underline{c}_0^{N-1} = \underline{u}_0^{N-1} \mathbf{G}_p$, where again $\mathbf{G}_p$ is the polar transform matrix.

With the above description, it can be seen that the two rate profilings that are specified by the index sets $\mathcal{A}$, $\mathcal{P}$ and $\mathcal{D}$, and the convolutional parameters (N_c, K_c) determine an MPAC code.

They also interplay in determining the MPAC codes' optimal and practical error-correction performances.

In particular, rate profiling II assigns the convolutional output and the remaining information bits based on the polarized subchannels reliabilities, while rate profiling I is further designed to provide a good weight distribution of MPAC codes. This effect will be revealed in Section IV. To realize this design, GA [10] and an iRMP ordering [13] are utilized. Selection of the convolutional parameters (N_c, K_c) will be further discussed in Sections III-C V.

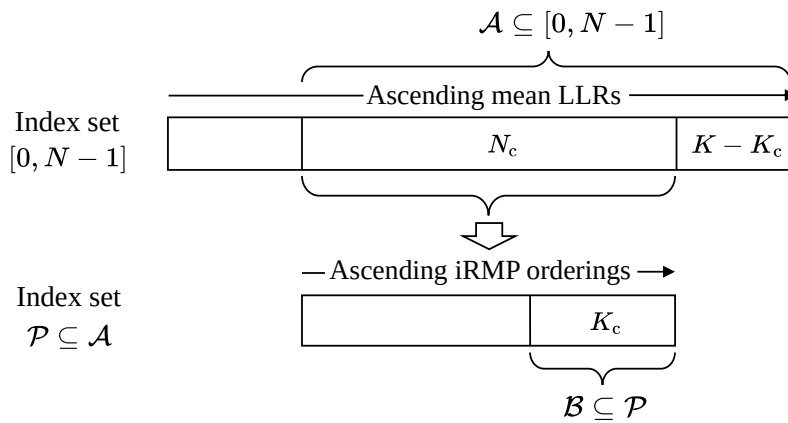


Fig. 3. Construction of index sets.

Fig. 3 shows the construction of index sets in the two rate profilings. It begins with the construction of index sets $\mathcal{A}$ and $\mathcal{P}$ of the rate profiling II using GA. By assuming an all-zero codeword, GA estimates the mean LLR value of all polarized subchannels $W_N^{(i)}$, which are

denoted as $\mathcal{L}(W_N^{(i)})$ and $i \in [0, N - 1]$. Note that the subchannel $W_N^{(i)}$ with a greater $\mathcal{L}(W_N^{(i)})$ is more reliable, and vice versa. Hence, the N subchannels can be ordered by their reliabilities, yielding a refreshed subchannel index tuple $(j_0, j_1, \dots, j_{N-1})_{\text{GA}}$, which implies

$$\mathcal{L}(W_N^{(j_0)}) > \mathcal{L}(W_N^{(j_1)}) > \dots > \mathcal{L}(W_N^{(j_{N-1})}).$$

Since $\mathcal{A}$ is the sorted index set of the $K - K_c + N_c$ most reliable subchannels, we have

$$\mathcal{A} = \{j_0, j_1, \dots, j_{K-K_c+N_c-1}\}. \quad (12)$$

Among $\mathcal{A}$, the index set of the N_c least reliable subchannels is

$$\mathcal{P} = \{j_{K-K_c}, j_{K-K_c+1}, \dots, j_{K-K_c+N_c-1}\}. \quad (13)$$

When considering the design of rate profiling I, we also use $\mathcal{I}$ and $\mathcal{F}$ to denote the *general information set* and *general frozen set* w.r.t. the information set and the frozen set of the inner polar code, respectively. For simplicity, subscripts $_c$ and $_p$ are used to denote those that will undergo the convolutional transform and those that will not, respectively. Hence, $\mathcal{I} = \mathcal{I}_c \cup \mathcal{I}_p$ and $\mathcal{F} = \mathcal{F}_c \cup \mathcal{F}_p$.

Consequently, rate profiling II constructs $\mathcal{I}_p = \mathcal{A} \setminus \mathcal{P}$, $\mathcal{F}_p = \mathcal{A}^c$ and $\mathcal{P} = \mathcal{I}_c \cup \mathcal{F}_c$ under the GA ordering, while rate profiling I only constructs $\mathcal{I}_c$ and $\mathcal{F}_c$.

Selecting the indices that indicate rows of higher weights in $\mathbf{G}_p$ in forming $\mathcal{I}$ would result in a better weight distribution for the code [12] [13]. For simplicity, let $r(i) = 2^{|\mathcal{S}_i|} = w(\mathbf{g}_i)$ denote the weight of row $\mathbf{g}_i$, where $i \in [0, N - 1]$ and $\mathcal{S}_i$ denotes the support of $\text{bin}(i)$. To improve the weight distribution of an MPAC code, a feasible approach is to construct $\mathcal{I}_c$ by selecting the indices with higher $r(i)$ values from the set $\mathcal{P}$. It can be realized by the iRMP rate profiling [13], as it can often result in a better weight distribution than the RM-polar rate profiling [12].

Definition 1 (iRMP Ordering). Given indices $i, j \in [0, N - 1]$, we define iRMP ordering $(i, j)_{\text{iRMP}}$ if they satisfy either of the following conditions:

- $r(i) > r(j)$;
- $r(i) = r(j)$ and $\mathcal{L}(W_N^{(i)}) \geq \mathcal{L}(W_N^{(j)})$.

Note that the mean LLR of subchannel $W_N^{(i)}$, i.e., $\mathcal{L}(W_N^{(i)})$, can be estimated by GA for $i \in [0, N - 1]$.

Since $\mathcal{L}(W_N^{(i)}) \neq \mathcal{L}(W_N^{(j)})$ for any $i, j \in [0, N - 1]$ and $i \neq j$, it is straightforward to verify

that the iRMP ordering is a total order. Note that any two indices belonging to $[0, N - 1]$ are comparable under the iRMP ordering.

Under the iRMP ordering, the indices in set $\mathcal{P}$ are ordered, yielding an N_c -tuple $(j_{[0]}, j_{[1]}, \dots, j_{[N_c-1]})_{\text{iRMP}}$ and $[\ast]$ is a mapping of $[0, N_c - 1] \rightarrow [K - K_c, K - K_c + N_c - 1]$. The tuple implies that for any $a', b' \in [0, N_c - 1]$ and $a' < b'$, we have $(j_{[a']}, j_{[b']})_{\text{iRMP}}$. Indices with the K_c highest iRMP orderings are selected to construct set $\mathcal{B}$ as

$$\mathcal{B} = \{j_{[0]}, j_{[1]}, \dots, j_{[K_c-1]}\}, \quad (14)$$

where $\mathcal{B} \subseteq \mathcal{P}$ and $|\mathcal{B}| = K_c$. Consequently, $\mathcal{I}_c = \mathcal{B}$ and $\mathcal{F}_c = \mathcal{P} \setminus \mathcal{B}$.

Index set $\mathcal{D}$ characterizes rate profiling I in disposing information bits $\underline{m}_0^{K_c-1}$, which is constructed based on sets $\mathcal{B}$ and $\mathcal{P}$ as follows. Elements of $\mathcal{B}$ are first represented in an ascending order over a vector, yielding $\underline{b}_0^{K_c-1} = (b_0, b_1, \dots, b_{K_c-1})$, where $b_0 < b_1 < \dots < b_{K_c-1}$. A mapping $F_{\mathcal{B}}(\ast)$ is further defined as $\mathcal{B} = \{b_i | i \in [0, K_c - 1]\} \rightarrow [0, K_c - 1]$. This implies that the index of the coordinate b_i is determined by $i = F_{\mathcal{B}}(b_i)$. Similarly, mapping $F_{\mathcal{P}}(\ast)$ is defined as $\mathcal{P} \rightarrow [0, N_c - 1]$. Subsequently, index set $\mathcal{D}$ can be further constructed as

$$\mathcal{D} = \{k \in [0, N_c - 1] \mid k = F_{\mathcal{P}}(b), b \in \mathcal{B}\}, \quad (15)$$

where $|\mathcal{D}| = K_c$. Since $\mathcal{P} = \mathcal{I}_c \cup \mathcal{F}_c$ and $\mathcal{B} = \mathcal{I}_c$, mapping functions $F_{\mathcal{I}_c \cup \mathcal{F}_c}(\ast)$ and $F_{\mathcal{I}_c}(\ast)$ are also defined, respectively. They are utilized to determine indices of the convolutional codeword $\underline{v}_0^{N_c-1}$ and the information bits $\underline{m}_0^{K_c-1}$ in the following HFSC decoding. Note that the above mentioned index sets and mapping functions are computed once offline for a particular (N, K) - (N_c, K_c) MPAC code.

B. HFSC Decoding

Fig. 4 shows block diagram of the HFSC decoding. This is a successive information recovery, where the Fano decoding and the SC decoding are deployed for estimating the message bits, based on the above mentioned index sets $\mathcal{I}_c$, $\mathcal{F}_c$, $\mathcal{I}_p$, and $\mathcal{F}_p$. In an MPAC code, only partial information bits undergo the convolutional transform. The HFSC decoding can subsequently limit the number of nodes that allow backward computation in Fano decoding, and rationalize the decoding complexity. It deploys the SC decoding to estimate the information bits $\underline{m}_{K_c}^{K-1}$ that are transmitted through the most reliable subchannels, i.e., those indexed by $\mathcal{I}_p$. Fano decoding

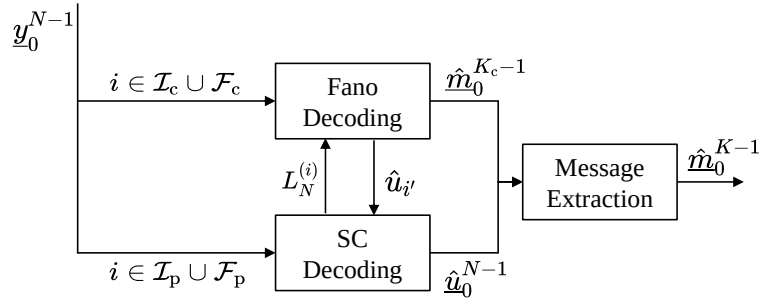


Fig. 4. Block diagram of the HFSC decoding.

will only be deployed to estimate the convolutional codeword $\underline{v}_0^{N_c-1}$ that are transmitted through the subchannels indexed by $\mathcal{I}_c \cup \mathcal{F}_c$. Meanwhile, it further estimates the information bits $\underline{m}_0^{K_c-1}$ that undergo the convolutional transform. Note that during the decoding, the decision LLRs are computed by the SC decoding rule [37]. The HFSC decoding operates as follows.

- For bits u_i that are indexed by $i \in \mathcal{I}_c \cup \mathcal{F}_c$, Fano decoding requests the SC rules for computing the decision LLRs $L_N^{(i)}$ and further determines the path metrics $\{M(\underline{\hat{u}}_0^i)\}_{\max}$ and $\{M(\underline{\hat{u}}_0^i)\}_{\min}$. Following a node visit as described in Section II-C, it estimates $\hat{u}_{i'}$, where $i' \leq i$. This indicates that the decoder can potentially correct the previously decoded bits through backward computation. Furthermore, if $i' \in \mathcal{I}_c$, the information bit $\hat{m}_j$ can be determined by the convolutional constraint of (9), where the index $j = F_{\mathcal{I}_c}(i')$. The Fano decoder feeds back the estimated bits $\hat{u}_{i'}$ to the SC decoder and continues the decoding.
- For bits u_i that are indexed by $i \in \mathcal{I}_p \cup \mathcal{F}_p$, the SC decoding estimates them based on its decision LLR $L_N^{(i)}$ as in (5). In particular, when $i \in \mathcal{F}_p$, it directly sets the frozen bits $\hat{u}_i = 0$. It also updates the path metric $M(\underline{\hat{u}}_0^i)$ based on the estimated bit $\hat{u}_i$.

Once $\hat{u}_{N-1}$ is estimated, or equivalently, when $i = N$, the information bits $\hat{m}_0^{K-1}$ can be retrieved from both $\hat{m}_0^{K_c-1}$ and $\hat{u}_0^{N-1}$. Algorithm 2 summarizes the HFSC decoding, where the input index sets $\mathcal{I}_c$, $\mathcal{F}_c$, $\mathcal{I}_p$ and $\mathcal{F}_p$ are determined by the rate profilings I and II.

Note that the decoding computation threshold can also be utilized for the HFSC decoding to limit the decoding latency and complexity, which will be introduced in the following subsection.

C. Simulation Results

We first study the performance-complexity tradeoff of the proposed MPAC codes numerically. Its theoretical insights and improved design will be presented in Sections IV. In this paper, all

Algorithm 2: The HFSC Decoding

Input: $\underline{y}_0^{N-1}$, $\mathcal{I}_c$, $\mathcal{F}_c$, $\mathcal{I}_p$, $\mathcal{F}_p$;
Output: $\hat{\underline{m}}_0^{K-1}$;

- 1 **Initialize** $i = 0$ and $j = 0$;
- 2 **While** $i \neq N$ **do**
 - 3 Compute decision LLR $L_N^{(i)}$ as in (4);
 - 4 **If** $i \in \mathcal{I}_c \cup \mathcal{F}_c$ **then** // Fano decoder
 - 5 Compute path metrics $M(\hat{\underline{u}}_1^i)$ as in (7);
 - 6 Perform a node visit of Fano decoding;
 - 7 Estimate $\hat{u}_{i'}$ with $i' \leq i$ and let $i = i' + 1$;
 - 8 **If** $i' \in \mathcal{I}_c$ **then**
 - 9 Let $j = F_{\mathcal{I}_c}(i')$ and determine $\hat{m}_j$ by convolutional constraint as in (9);
 - 10 **Else** // SC decoder
 - 11 **If** $i \in \mathcal{F}_p$ **then** set $\hat{u}_i = 0$;
 - 12 **If** $i \in \mathcal{I}_p$ **then** // SC decoding
 - 13 Estimate $\hat{u}_i$ based on $L_N^{(i)}$ as in (5)
 - 14 Compute $M(\hat{\underline{u}}_0^i)$ as in (7) and let $i = i + 1$;
- 15 Retrieve $\hat{\underline{m}}_0^{K-1}$ from $\hat{\underline{m}}_0^{K_c-1}$ and $\hat{\underline{u}}_0^{N-1}$;
- 16 **Return** $\hat{\underline{m}}_0^{K-1}$

codes and decoding performances are evaluated over the AWGN channel using BPSK modulation. The proposed schemes are compared with relevant coding schemes, including polar codes under SC decoding, PAC codes under Fano decoding, and CRC-polar codes under SCL decoding. Both the MPAC codes and PAC codes employ the convolutional generator sequence $(1, 0, 1, 1, 0, 1, 1)$. MPAC codes are designed using GA at an SNR per codeword symbol of $E_s/N_0 = 0$ dB, while PAC codes are constructed using the RM rate profiling [7]. For both the HFSC decoding and Fano decoding, the step size $\Delta = 2$ which is numerically chosen in optimizing performance-complexity tradeoff [22]. Both the CRC-polar codes and polar codes are designed using the reliability sequence in 5G standard [38]. Given N and K , an (N, K) -(N_c, K_c) MPAC code is alternatively denoted as MPAC-(N_c, K_c). The convolutional parameters (N_c, K_c) will be determined by maximizing the HFSC decoding performance, which helps further verify the performance-complexity tradeoff advantage of the MPAC codes.

Decoding complexity is measured as the average number of LLR computations required to decode a codeword. A decoding LLR computation threshold Φ is introduced to bound the decoding complexity and latency of both the HFSC decoding and Fano decoding. It is a

multiplicity of the SC decoding complexity, i.e., $\Phi = \eta N \log_2 N$, where $\eta \in \mathbb{N}^+$ is referred as the normalized complexity factor. For both the HFSC decoding and Fano decoding, once Φ is reached, the SC decoding will be switched to in estimating the remaining information bits. This prevents the decoding from lingering over the decoding tree, rationalizing the otherwise unlimited decoding computation.

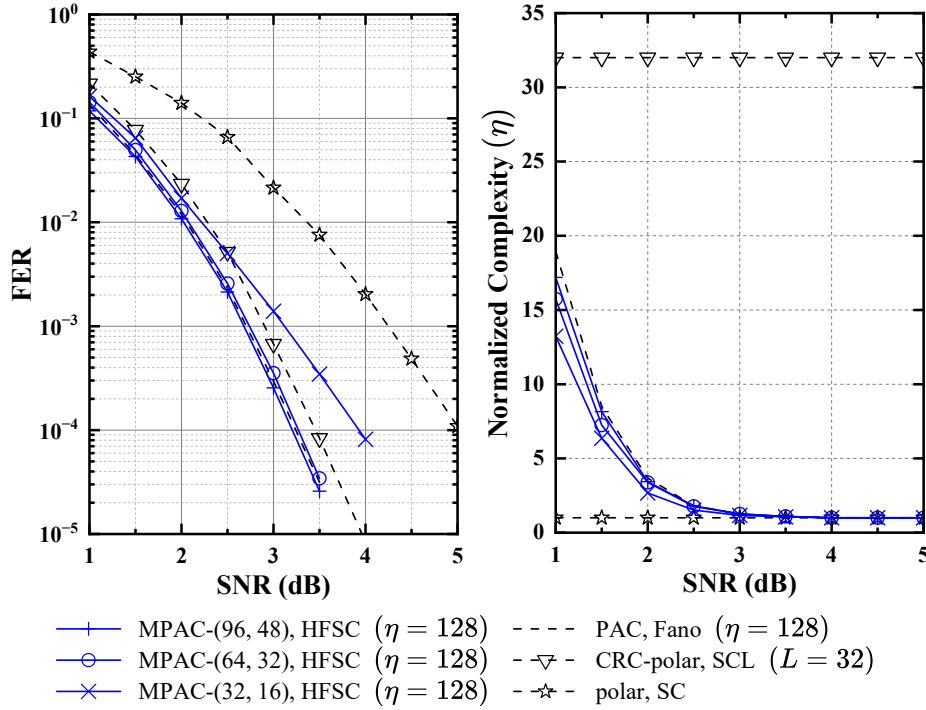


Fig. 5. Performance comparison of MPAC codes, PAC codes, polar codes and CRC-polar codes, where $N = 128$ and $K = 64$.

Fig. 5 shows the decoding frame error rate (FER) and complexity performance of different coding schemes with $N = 128$ and $K = 64$. For yielding a good FER performance under the HFSC decoding, optional parameters $(N_c, K_c) = (96, 48)$, $(64, 32)$ and $(32, 16)$ are chosen for MPAC codes, which are obtained from our numerical study in [35]. A decoding computation threshold of $\eta = 128$ is also provided for the HFSC decoding and Fano decoding as in [35]. It can be seen that the MPAC-(96, 48) code can slightly outperform the (128, 64) PAC code. Furthermore, when compared with the CRC-polar code, the MPAC-(96, 48) code achieves 0.25 dB power gain at the FER of 10^{-4} with a smaller complexity. It can also be seen that the MPAC-(64, 32) code performs nearly the same as the (128, 64) PAC code, but with a lower decoding complexity. Moreover, both the MPAC-(96, 48) code and MPAC-(64, 32) code can outperform

the MPAC-(32, 16) code, validating the effectiveness of the convolutional parameters which are listed in [35]. It implies that a sufficient portion of information bits needs to undergo convolutional transform to achieve a good decoding performance.

IV. MWD ANALYSIS

This section analyzes the MWD of MPAC codes, including the formation of the MWCs and their coset-wise enumeration. Compared with the existing SCL-based enumeration [34], it has a much lower computational complexity. More importantly, this analysis provides a new metric for designing MPAC codes, especially in determining the convolutional transform parameters (N_c, K_c) . We first revisit the formation of MWCs for polar codes. By considering the impact of convolutional transform and frozen bits, we present the general characterization of row combinations of the MWCs in MPAC codes together with the validation approach. Based on this, an efficient algorithm is proposed to determine the number of MWCs.

A. Formation of the MWCs for Polar Codes

The set theoretical approach [20] in analyzing the MWD of polar codes is applied to show the intrinsic structure to the proposed MPAC codes.

By enumerating all the combinations, the formation and the number of MWCs can be obtained. The approach can be used to analyze polar like codes with the partial order property, which will be defined as follows.

Definition II (Partial Order [39]). Given subchannel indices $i, j \in [0, N - 1]$, the partial order $i \preceq j$ holds if they satisfy one of the three conditions:

- $\mathcal{S}_i \subseteq \mathcal{S}_j$;
- $\mathcal{S}_j = (\mathcal{S}_i \setminus \{a\}) \cup \{b\}$ for a pair of $a \in \mathcal{S}_i, b \notin \mathcal{S}_i$ and $a < b$;
- There exists an index $k \in [0, N - 1]$ satisfying $i \preceq k$ and $k \preceq j$.

Note that there may exist some pairs of indices that neither of the index precedes the other under the partial order. For this, the following Definition III is needed.

Definition III (Partial Order Property [20]). A set $\mathcal{I} \subseteq [0, N - 1]$ is said to satisfy the partial order property if none of the index pair $i \in \mathcal{I}, i^c \in \mathcal{I}^c$ satisfies the partial order $i \preceq i^c$. Or equivalently, we have $i \not\preceq i^c$ for any indices $i \in \mathcal{I}$ and $i^c \in \mathcal{I}^c$.

With the above definition, if a set $\mathcal{I}$ satisfies the partial order property, it can be found that all the indices in $\mathcal{I}$ compose a chain of partial orders. Moreover, for every pair of indices $i \in \mathcal{I}$ and $j \in [0, N - 1]$, if $i \preceq j$, $j \in \mathcal{I}$.

Given a polar code $\mathcal{C}(\mathcal{I})$ with information set $\mathcal{I}$ that satisfies the partial order property, the analysis of its MWCs starts from dividing the code $\mathcal{C}(\mathcal{I})$ into $|\mathcal{I}|$ disjoint cosets.

Definition IV (Polar Coset [20]). Given an N -dimensional polar transform $\mathbf{G}_p$ and an information set $\mathcal{I} \in [0, N - 1]$, the coset is defined by

$$\mathcal{C}_i(\mathcal{I}) = \left\{ \mathbf{g}_i \oplus \bigoplus_{h \in \mathcal{H}} \mathbf{g}_h \mid \mathcal{H} \subseteq \mathcal{I} \setminus [0, i] \right\}, \quad (16)$$

where the i -th row of $\mathbf{G}_p$, i.e., $\mathbf{g}_i$, is the coset leader.

In other words, coset $\mathcal{C}_i(\mathcal{I})$ is the set of codewords generated by $\mathcal{C}(\mathcal{I} \setminus [0, i - 1])$, where $\mathcal{C}(\mathcal{I} \setminus [0, i - 1])$ is the subcode of polar code $\mathcal{C}(\mathcal{I})$. The following Lemma 1 indicates that the Hamming weight of coset leader $\mathbf{g}_i$, where $i \in \mathcal{I}$ is a lower bound for the weight of any codeword in the coset $\mathcal{C}_i(\mathcal{I})$.

Lemma 1 ([20], [40]). For any $i \in [0, N - 1]$ and $\mathcal{H} \subseteq [i + 1, N - 1]$, we have

$$w(\mathbf{g}_i \oplus \bigoplus_{h \in \mathcal{H}} \mathbf{g}_h) \geq w(\mathbf{g}_i). \quad (17)$$

By comparing the row weights of all coset leaders, the minimum distance of the polar code $\mathcal{C}(\mathcal{I})$ can be obtained by

$$\begin{aligned} d_{\min} &= \omega_{\min} \\ &= \min(\{w(\mathbf{g}_i) \mid i \in \mathcal{I}\}). \end{aligned} \quad (18)$$

Note that the weight of row $\mathbf{g}_i$ can be easily calculated by $w(\mathbf{g}_i) = 2^{|\mathcal{S}_i|}$, where $i \in [0, N - 1]$. Further let $A_{\omega_{\min}}(\mathcal{I})$ and $A_{i, \omega_{\min}}(\mathcal{I})$ denote the number of MWCs in polar code $\mathcal{C}(\mathcal{I})$ and coset $\mathcal{C}_i(\mathcal{I})$, respectively. Based on Lemma 1, we have $A_{i, \omega_{\min}}(\mathcal{I}) = 0$ for every index $i \in \{j \in \mathcal{I} \mid w(\mathbf{g}_j) > \omega_{\min}\}$. Consequently, only the cosets with their leaders' weight being equal to $\omega_{\min}$ need to be considered. All linear combinations that form the MWCs in coset $\mathcal{C}_i(\mathcal{I})$, where $i \in \mathcal{I}$ and $w(\mathbf{g}_i) = \omega_{\min}$, can be explicitly characterized by the following Lemma 2.

Lemma 2 ([20], [32]). Given an information set $\mathcal{I} \subseteq [0, N - 1]$ that satisfies the partial order property and coset leader index $i \in \mathcal{I}$ with $w(\mathbf{g}_i) = \omega_{\min}$, the MWCs in polar coset $\mathcal{C}_i(\mathcal{I})$ can

be formed by rows as

$$w(\mathbf{g}_i \oplus \bigoplus_{j \in \mathcal{J}} \mathbf{g}_j \oplus \bigoplus_{m \in \mathcal{M}(\mathcal{J})} \mathbf{g}_m) = \omega_{\min}, \quad (19)$$

where the rows with index defined in sets $\mathcal{J} \subseteq \mathcal{K}_i$ and $\mathcal{M}(\mathcal{J})$ are called the core rows and the balancing rows, respectively. Moreover, the sets $\mathcal{K}_i$ and $\mathcal{M}(\mathcal{J})$ can be constructed by the index of i and the so called $\mathcal{M}$ -construction, respectively.

For an index $i \in [0, N - 1]$, the set $\mathcal{K}_i$ captures every index $j \in [i + 1, N - 1]$ that satisfies $w(\mathbf{g}_j) \geq w(\mathbf{g}_i \oplus \mathbf{g}_j) = w(\mathbf{g}_i)$. Equivalently, set $\mathcal{K}_i$ is defined as

$$\mathcal{K}_i \triangleq \{j \in [i + 1, N - 1] \mid |\mathcal{S}_j \setminus \mathcal{S}_i| = 1\}. \quad (20)$$

This set can be determined by applying the *addition* operation and the *left swap* operation on binary vector $\text{bin}(i)$ as in [20].

Given a set $\mathcal{J} \subseteq \mathcal{K}_i$, the corresponding balancing set $\mathcal{M}(\mathcal{J})$ can be constructed using the $\mathcal{M}$ -construction [20]. However, the $\mathcal{M}$ -construction is only conducted when all the indices of set $\mathcal{J}$ are known. It is more efficient to construct set $\mathcal{M}(\mathcal{J})$ through an ongoing approach [41] which enlists the indices of set $\mathcal{J}$ one-by-one.

The partial order property of information set $\mathcal{I}$ promises both sets $\mathcal{K}_i$ and $\mathcal{M}(\mathcal{J})$ are disjoint subsets of $\mathcal{I} \cap [i + 1, N - 1]$. That says $\mathcal{K}_i \subseteq \mathcal{I} \cap [i + 1, N - 1]$, $\mathcal{M}(\mathcal{J}) \subseteq \mathcal{I} \cap [i + 1, N - 1]$ and $\mathcal{K}_i \cap \mathcal{M}(\mathcal{J}) = \emptyset$.

As a result, the value of $\underline{u}_0^{N-1}$ can be controlled based on

$$u_x = \begin{cases} 1, & \text{if } x \in \{i\} \cup \mathcal{J} \cup \mathcal{M}(\mathcal{J}); \\ 0, & \text{if } x \in [0, N - 1] \setminus (\{i\} \cup \mathcal{J} \cup \mathcal{M}(\mathcal{J})). \end{cases} \quad (21)$$

The corresponding MWCs can be further obtained by polar encoding as in (3).

Consequently, the number of MWCs in $\mathcal{C}(\mathcal{I})$ can be calculated as defined by the following Corollary 3.

Corollary 3 ([20]). Given an information set $\mathcal{I} \subseteq [0, N - 1]$ that satisfies the partial order property, the number of MWCs of $\mathcal{C}_i(\mathcal{I})$ is

$$A_{\omega_{\min}}(\mathcal{I}) = \sum_{i \in \mathcal{I}: w(\mathbf{g}_i) = \omega_{\min}} 2^{|\mathcal{K}_i|}. \quad (22)$$

B. Formation of the MWCs for MPAC Codes

The above mentioned characterization of row combinations of MWCs for polar codes shows that it is important to define the index sets that correspond to the support of vector $\underline{u}_0^{N-1}$. Therefore, in forming the MWCs of MPAC codes, the index sets $\mathcal{I}_c$, $\mathcal{F}_c$, $\mathcal{I}_p$ and $\mathcal{F}_p$ shall be characterized. For our description convenience, they are also illustrated in Fig. 6.

The general information set $\mathcal{I} = \mathcal{I}_c \cup \mathcal{I}_p$ and frozen set $\mathcal{F} = \mathcal{F}_c \cup \mathcal{F}_p$ are further obtained. The rate profiling I ensures the subchannel indexes i with higher $w(\mathbf{g}_i)$ are chosen, yielding a higher minimum row weights, i.e., $\min(\{w(\mathbf{g}_i) \mid i \in \mathcal{I}\})$, for MPAC codes.

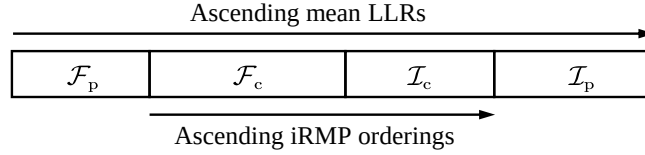


Fig. 6. Index sets of N subchannels under MPAC coding.

It is worth mentioning that the method proposed for MWCs enumeration of MPAC code can be realized through checking the legality of every MWC enumeration. As a result, their MWCs can be formed by (19) and further be enumerated by the row combination manner as (21). In other words, we need to set $u_x = 1$ for every $x \in \{i\} \cup \mathcal{J} \cup \mathcal{M}(\mathcal{J})$. However, it has been observed in [33] that the outer convolutional transform with the frozen bits as inputs generates constraint on the values of some u_x . In the proposed MPAC codes, $\underline{u}_0^{N-1} = (\underline{u}_{\mathcal{I}_c \cup \mathcal{F}_c}, \underline{u}_{\mathcal{I}_p}, \underline{u}_{\mathcal{F}_p})$, where the values of $\underline{u}_{\mathcal{I}_c \cup \mathcal{F}_c}$, $\underline{u}_{\mathcal{I}_p}$ and $\underline{u}_{\mathcal{F}_p}$ are the output vector of convolutional transform, information bits and frozen bits, respectively. In the following, we will discuss the influence of convolutional transform and the frozen bits on the value of u_x , which are characterized in three cases.

Case I: When $x \in \mathcal{I}_c \cup \mathcal{F}_c$, u_x is the convolutional output, which is determined by $u_x = \bigoplus_{j=0}^t g_j s_{k-j}$, or equivalently, by applying $\text{conv1b}(\cdot)$, where $k = F_{\mathcal{I}_c \cup \mathcal{F}_c}(x)$. Considering the constraint of t previous convolutional inputs, we have $u_x = V \oplus s_k$, where $V = \bigoplus_{j=1}^t g_j s_{k-j}$. Since the value of s_k is determined by either an information bit or a frozen bit, we consider the corresponding subcases of $x \in \mathcal{I}_c$ and $x \in \mathcal{F}_c$.

Case I-A: When $x \in \mathcal{I}_c$, $s_k = 0$ or 1 . Hence, we can obtain $u_x = 1$ by choosing s_k appropriately such that

$$s_k = \begin{cases} 1, & \text{if } V = 0; \\ 0, & \text{if } V = 1. \end{cases} \quad (23)$$

In obtaining $u_x = 0$, we can take the opposites of (23).

Case I-B: When $x \in \mathcal{F}_c$, s_k is determined by a frozen bit, and let $s_k = 0$. Consequently, u_x is only determined by the previous input constraint, i.e., $u_x = V$. If the previous inputs cause $V = 1$, we have $u_x = 1$. Otherwise, $u_x = 0$.

Case II: When $x \in \mathcal{I}_p$, u_x is determined by an information bit, $u_x = 0$ or 1 .

Case III: When $x \in \mathcal{F}_p$, u_x is directly determined by a frozen bit and $u_x = 0$.

It can be seen that in *Case I-A* and *Case I-B*, we have no control on the value of u_x . Based on the setting of $\underline{u}_0^{N-1}$ in obtaining the MWCs as in (21), we focus on the value of u_x that is only determined by convolutional operation, where $x \in (\mathcal{F}_c \cup \mathcal{F}_p) \cap [i+1, N-1]$. Since in **Case III** $u_x = 0$, we turn to focus on **Case I-B**. Consequently, the index set $\tilde{\mathcal{F}}_c$ is defined as

$$\tilde{\mathcal{F}}_c \triangleq \{f \in \mathcal{F}_c \cap [i+1, N-1] | u_f = 1\}. \quad (24)$$

The generation of $\tilde{\mathcal{F}}_c$ implies that another term $\bigoplus_{f \in \tilde{\mathcal{F}}_c} \mathbf{g}_f$ will be included in the row combination of (19). A subset of $\tilde{\mathcal{F}}_c$ is further defined as

$$\mathcal{F}_{\mathcal{J}} = \{f \in \tilde{\mathcal{F}}_c \mid |\mathcal{S}_f \setminus \mathcal{S}_i| = 1\}. \quad (25)$$

Remark 4. The inclusion of the additional term $\bigoplus_{f \in \tilde{\mathcal{F}}_c} \mathbf{g}_f$ in the MWC row combination of (19) does not reduce the minimum distance of an MPAC code. This is due to the fact that for any $f \in \tilde{\mathcal{F}}_c$, the enlisted row $\mathbf{g}_f$ is not the coset leader of any coset in the code. Based on Lemma 1, since the general information set $\mathcal{I} = \mathcal{I}_c \cup \mathcal{I}_p$ remains unchanged, the minimum distance of the MPAC code is maintained as

$$\begin{aligned} d_{\min} &= \omega_{\min} \\ &= \min(\{w(\mathbf{g}_i) \mid i \in \mathcal{I} \text{ and } \mathcal{I} = \mathcal{I}_c \cup \mathcal{I}_p\}). \end{aligned} \quad (26)$$

Please note that the above property does not always hold for all MPAC codes. But in this work, we only consider the MPAC codes that yield (26). Theorem 5 further characterizes the more general row combination in obtaining the MWCs of an MPAC code.

Theorem 5. Given an MPAC code that is defined by the subchannel index sets $\mathcal{I}_c$, $\mathcal{F}_c$, $\mathcal{I}_p$ and $\mathcal{F}_p$ where index i is chosen such that $w(\mathbf{g}_i) = \omega_{\min}$, the minimum weight codewords in coset

$\mathcal{C}_i(\mathcal{I})$ can be further redefined based on (25) as

$$w(\mathbf{g}_i \oplus \bigoplus_{j \in \mathcal{J} \cup \mathcal{F}_{\mathcal{J}}} \mathbf{g}_j \oplus \bigoplus_{m \in \mathcal{M}(\mathcal{J} \cup \mathcal{F}_{\mathcal{J}})} \mathbf{g}_m) = \omega_{\min}, \quad (27)$$

where $\mathcal{J} \subseteq \mathcal{K}_i$ and $\mathcal{M}(\mathcal{J})$ are constructed as in Lemma 2. The set $\mathcal{F}_{\mathcal{J}}$ can be represented based on (24) and (25) as

$$\mathcal{F}_{\mathcal{J}} = \{f \in \mathcal{F}_c \cap [i+1, N-1] \mid u_f = 1 \text{ and } |\mathcal{S}_f \setminus \mathcal{S}_i| = 1\}.$$

Proof: It is a generalization of Lemma 2.

The main difference between MWC generation of polar codes and that of MPAC codes lies in the index set $\mathcal{F}_{\mathcal{J}}$. It is caused by the convolutional constraints and the frozen bits as in **Case I-B**. Furthermore, since the indices of set $\mathcal{F}_{\mathcal{J}}$ have the same property as those of the set $\mathcal{J} \subseteq \mathcal{K}_i$, we can replace the sets $\mathcal{J}$ and $\mathcal{M}(\mathcal{J})$ of (19) by $\mathcal{J} \cup \mathcal{F}_{\mathcal{J}}$ and $\mathcal{M}(\mathcal{J} \cup \mathcal{F}_{\mathcal{J}})$, respectively. This immediately leads to (27).

In the following subsection IV-C, we will propose an efficient approach to yield sets $\mathcal{J}$ and $\mathcal{F}_{\mathcal{J}}$, which can be enumerated one by one as in **Algorithm 3** $\square$

Fig. 7 illustrates Venn diagram of the associated index sets in characterizing the MWCs of MPAC codes. In particular, the index sets of rows that form the MWCs are denoted by the red boxes. We further combine these index sets into the set

$$\mathcal{U}_i(\mathcal{J}) = \{i\} \cup (\mathcal{J} \cup \mathcal{F}_{\mathcal{J}}) \cup \mathcal{M}(\mathcal{J} \cup \mathcal{F}_{\mathcal{J}}), \quad (28)$$

such that in obtaining the MWC, it should be

$$u_x = \begin{cases} 1 & \text{if } x \in \mathcal{U}_i(\mathcal{J}); \\ 0 & \text{if } x \in [0, N-1] \setminus \mathcal{U}_i(\mathcal{J}). \end{cases} \quad (29)$$

Therefore, the number of MWCs of the proposed MPAC codes can be determined. However, as discussed in **Case I-B** and **Case III**, for $f \in \mathcal{F}_c \cup \mathcal{F}_p$, we have no control on the value of u_f . If eq. (29) cannot be ensured, the corresponding MWC cannot be formed.

The following subsection introduces the verification of the MWCs for MPAC codes and determines their exact number.

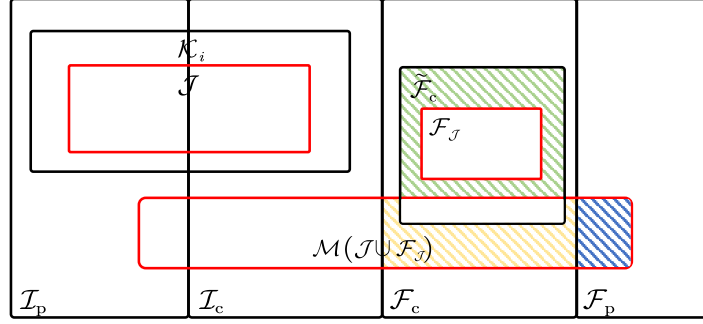


Fig. 7. Venn diagram of the associated index sets for indices $[i + 1, N - 1]$.

C. Verification of the MWCs for MPAC Codes

Based on Corollary 3, the number of MWCs characterized in (27) can be obtained by enumerating all the possible index sets $\mathcal{U}_i(\mathcal{J})$ that are defined as in (28).

Since the index set $\mathcal{M}(\mathcal{J} \cup \mathcal{F}_{\mathcal{J}})$ is constructed based on set $\mathcal{J} \cup \mathcal{F}_{\mathcal{J}}$, all possible index sets $\mathcal{J} \cup \mathcal{F}_{\mathcal{J}}$ need to be found [33].

Lemma 6. The number of all possible index sets $\mathcal{J} \cup \mathcal{F}_{\mathcal{J}}$ is equal to $2^{|\mathcal{K}_i|}$.

Proof: The number of all possible index sets $\mathcal{J} \subseteq \mathcal{K}_i$ is $2^{|\mathcal{K}_i|}$. Since set $\mathcal{F}_{\mathcal{J}}$ is generated based on the convolutional outputs, it is related to the previous convolutional inputs. When obtaining the MWCs of an MPAC code, the convolutional inputs are determined by (29). In other words, $\mathcal{F}_{\mathcal{J}}$ depends on set $\mathcal{J}$. Therefore, the number of all possible sets $\mathcal{J} \cup \mathcal{F}_{\mathcal{J}}$ is equal to that of all possible sets $\mathcal{J}$, i.e., $2^{|\mathcal{K}_i|}$. $\square$

Combining both **Case I-B** and **Case III** of Section IV-B, $u_x = \bigoplus_{j=1}^t g_j s_{k-j} = V$ and $k = F_{\mathcal{I}_c \cup \mathcal{F}_c}(x)$ for $x \in \mathcal{F}_c$ and $u_x = 0$ for $x \in \mathcal{F}_p$, respectively. Consequently, for some $\mathcal{J} \subseteq \mathcal{K}_i$, the value of u_x and $x \in \mathcal{F}_c \cup \mathcal{F}_p$, cannot be chosen to ensure $\text{supp}(\underline{u}_0^{N-1}) = \mathcal{U}_i(\mathcal{J})$. This results in some codewords that are formed as (27) cannot be generated by the MPAC encoding. For better representation, let us further define

$$\mathcal{E}_{\mathcal{J}}^1 = \{(\tilde{\mathcal{F}}_c \setminus \mathcal{F}_{\mathcal{J}}) \setminus ((\mathcal{F}_c \cap [i + 1, N - 1]) \cap \mathcal{M}(\mathcal{J} \cup \mathcal{F}_{\mathcal{J}}))\},$$

$$\mathcal{E}_{\mathcal{J}}^2 = \{((\mathcal{F}_c \cap [i + 1, N - 1]) \cap \mathcal{M}(\mathcal{J} \cup \mathcal{F}_{\mathcal{J}})) \setminus \tilde{\mathcal{F}}_c$$

$$\mathcal{E}_{\mathcal{J}}^3 = \{(\mathcal{F}_p \cap [i + 1, N - 1]) \cap \mathcal{M}(\mathcal{J} \cup \mathcal{F}_{\mathcal{J}})\}.$$

For a more illustrative presentation, the index sets $\mathcal{E}_{\mathcal{J}}^1$, $\mathcal{E}_{\mathcal{J}}^2$ and $\mathcal{E}_{\mathcal{J}}^3$ are marked by the green, yellow and blue shades in Fig 7, respectively. Considering the contradiction of $\text{supp}(\underline{u}_0^{N-1}) \neq \mathcal{U}_i(\mathcal{J})$, there are three criteria that can be utilized to validate the case in which the characterized codewords cannot be generated. For every $\mathcal{J} \subseteq \mathcal{K}_i$, when $x \in \mathcal{F}_c \cup \mathcal{F}_p$, the value of u_x needs to be checked as follows.

Criterion 1 (C1): $u_x = 1$, but $x \in \mathcal{E}_{\mathcal{J}}^1$;

Criterion 2 (C2): $u_x = 0$, but $x \in \mathcal{E}_{\mathcal{J}}^2$;

Criterion 3 (C3): $u_x = 0$, but $x \in \mathcal{E}_{\mathcal{J}}^3$.

In **C1**, since $x \in \mathcal{E}_{\mathcal{J}}^1$, u_x should be set as 0 such that (29) is met. However, we get $u_x = 1$ under the constraints of **Case I-B**, which implies the undesired row $\mathbf{g}_x$ has been included in the row combinations of (27). It prevents formation of the corresponding MWC. In **C2**, since $x \in \mathcal{E}_{\mathcal{J}}^2$, u_x should be set as 1. However, under the same constraints in **Case I-B**, we get $u_x = 0$, resulting in the desired row $\mathbf{g}_x$ being excluded from the row combinations of (27). **C3** also checks the case in which the characterized MWCs cannot be generated due to the absence of some specific rows. Note that a missing row is caused by the constraint of the corresponding frozen bit in **Case III** and its index $x \in \mathcal{E}_{\mathcal{J}}^3$. With the above three criteria, the exact number of the MWCs of an MPAC code is characterized as in Theorem 7.

Theorem 7. With the subchannel index sets $\mathcal{I}_c$, $\mathcal{F}_c$, $\mathcal{I}_p$ and $\mathcal{F}_p$ of MPAC codes, the number of MWCs in coset $\mathcal{C}_i(\mathcal{I})$ of the MPAC code is

$$A_{i, \omega_{\min}}^{\text{MPAC}} = 2^{|\mathcal{K}_i|} - X, \quad (30)$$

where X is the number of the MWCs that are characterized in Theorem 5, but cannot be formed. It is called the MWC number deficit.

Proof: Based on Lemma 6, the number of MWCs in $\mathcal{C}_i(\mathcal{I})$ of an MPAC code has an upper bound of $2^{|\mathcal{K}_i|}$. The upper bound is achieved only if for all $\mathcal{J} \subseteq \mathcal{K}_i$, the corresponding vector $\underline{u}_0^{N-1}$ satisfies $\text{supp}(\underline{u}_0^{N-1}) = \mathcal{U}_i(\mathcal{J})$. Assume X is the number of the sets $\mathcal{J}$ that $\text{supp}(\underline{u}_0^{N-1}) \neq \mathcal{U}_i(\mathcal{J})$, we have $A_{i, \omega_{\min}}^{\text{MPAC}} = 2^{|\mathcal{K}_i|} - X$, where $X \in [0, 2^{|\mathcal{K}_i|}]$. The value of X is obtained by assessing the above mentioned criteria **C1**, **C2** and **C3**. $\square$

Recalling (24), (25) and the $\mathcal{M}$ -construction, for every index $f \in \tilde{\mathcal{F}}_c \setminus \mathcal{F}_{\mathcal{J}}$,

$\mathcal{M}(\mathcal{J} \cup \mathcal{F}_{\mathcal{J}})$ has the same property of $|\mathcal{S}_f \setminus \mathcal{S}_i| > 1$. Let us further define set

$$\mathcal{F}^* = \{f \in (\mathcal{F}_c \cup \mathcal{F}_p) \cap [i+1, N-1] \mid |\mathcal{S}_f \setminus \mathcal{S}_i| > 1\},$$

where $\mathcal{E}_{\mathcal{J}}^1 \cup \mathcal{E}_{\mathcal{J}}^2 \cup \mathcal{E}_{\mathcal{J}}^3 \subseteq \mathcal{F}^*$. The following Remark 8 reveals that if $\mathcal{F}^* = \emptyset$, the value of X can be set as 0 without enumerating the $2^{|\mathcal{K}_i|}$ characterized MWCs.

Remark 8. Since $\mathcal{E}_{\mathcal{J}}^1 \cup \mathcal{E}_{\mathcal{J}}^2 \cup \mathcal{E}_{\mathcal{J}}^3 \subseteq \mathcal{F}^*$, if one of the criteria **C1**, **C2** and **C3** is met, there exists at least an index f that satisfies $f \in \mathcal{F}^*$. If $\mathcal{F}^* = \emptyset$, none of the three criteria can be met and $X = 0$. Note that $\mathcal{F}^* = \emptyset$ if and only if $\{(\mathcal{F}_c \cup \mathcal{F}_p) \cap [i+1, N-1]\} = \emptyset$ or $|\mathcal{S}_f \setminus \mathcal{S}_i| = 1$ for all $f \in \{(\mathcal{F}_c \cup \mathcal{F}_p) \cap [i+1, N-1]\}$. The former case implies that every set $\mathcal{U}_i(\mathcal{J})$ is a subset of $\{(\mathcal{I}_c \cup \mathcal{I}_p) \cap [i, N-1]\}$, which is attainable. For the latter, $\mathcal{E}_{\mathcal{J}}^1 = \mathcal{E}_{\mathcal{J}}^2 = \mathcal{E}_{\mathcal{J}}^3 = \emptyset$. None of the above cases is consistent with **C1**, **C2** and **C3**. Hence, $X = 0$.

Except the above mentioned case, counting the number of MWCs in coset $\mathcal{C}_i(\mathcal{I})$, i.e., $A_{i, \omega_{\min}}^{\text{MPAC}}$, can be performed through enumerating all the $2^{|\mathcal{K}_i|}$ index sets $\mathcal{U}_i(\mathcal{J})$ and checking the three criteria for related vectors $\underline{u}_0^{N-1}$. Based on Theorem 7, we have $A_{i, \omega_{\min}}^{\text{MPAC}} = 2^{|\mathcal{K}_i|} - X$. An ongoing approach is applied to construct $\mathcal{U}_i(\mathcal{J})$. The values of u_x with $x \in [i, N-1]$ can then be checked such that the values of X and $A_{i, \omega_{\min}}^{\text{MPAC}}$ can be determined. This is summarized as in **Algorithm 3**, which evolves from Algorithm 1 of [41]. However, when focusing on the MPAC codes, the additional cases should be taken into account. During the bit-by-bit acquisition of u_x for each $\mathcal{J} \subseteq \mathcal{K}_i$, we only record the updates of the shift register states S . Notably, state S is updated only at bits u_x such that $x \in \mathcal{I}_c \cup \mathcal{F}_c$. The ongoing approach initializes $X = 0$ and proceeds to enumerate u_x for $x = i$ to $N-1$. According to the reached index of x , there are three cases summarized as follows.

- When reaching an index $x \in \mathcal{I}_c$ or $x \in \mathcal{I}_p$, if $x \in \mathcal{J}$, let $u_x = 1$ and partially construct $\mathcal{M}(\mathcal{J} \cup \mathcal{F}_{\mathcal{J}})$ that is denoted as $\tilde{\mathcal{M}}$, using the subroutine `addToM( $\cdot$ )` that was defined in Algorithm 2 of [41]. Index x is also needed to add to the partially formed $\mathcal{J}$, which is denoted as $\tilde{\mathcal{J}}$. In particular, when $x \in \mathcal{I}_c$, the value of u_x can be determined based on (29). Furthermore, the input bit s_k is determined by (9) and the state S is updated by `conv1b( $s_k, S, g_0^t$ )`, where $k = F_{\mathcal{I}_c \cup \mathcal{F}_c}(x)$.
- When reaching an index $x \in \mathcal{F}_c$ as in **Case I-B**, u_x is determined by the previous convolutional inputs, i.e., $[u_x, S] \leftarrow \text{conv1b}(0, S, g_0^t)$. If either criteria **C1** or **C2** is satisfied, X is increased by one. In particular, if $x \in \mathcal{F}_{\mathcal{J}}$, the operations for index $x \in \mathcal{J}$ will be

also taken.

It should be mentioned that the subroutine `addToM(·)` does not distinguish the indices in sets $\mathcal{J}$ and $\mathcal{F}_{\mathcal{J}}$.

- When reaching an index $x \in \mathcal{F}_{\mathcal{P}}$ as in **Case III**, u_x is frozen as 0. If criterion **C3** is satisfied, X is also increased by one.

With the above mentioned procedure, X can be determined and so can $A_{i,\omega_{\min}}^{\text{MPAC}}$ be obtained. For every $\mathcal{C}_i(\mathcal{I})$, complexity of the coset-wise enumeration is $O(2^{|\mathcal{K}_i|} \cdot (N - i))$. Moreover, by introducing a factor $\zeta = \max\{\mathcal{F}^*\}$, the computational complexity can be reduced to $O(2^{|\mathcal{K}_i^\zeta|} \cdot (\zeta - i))$. It follows a similar strategy that was introduced in [41]. Based on **Algorithm 3**, the set $\mathcal{K}_i$ (in line 4 of **Algorithm 3**) and the vector $\underline{u}_i^{N-1}$ (in line 10 of **Algorithm 3**) degenerate into set $\mathcal{K}_i^\zeta = \{j \in \mathcal{K}_i \mid j < \zeta\}$ and vector $\underline{u}_i^\zeta$, respectively. As a result, $A_{i,\omega_{\min}}^{\text{MPAC}} = 2^{|\mathcal{K}_i \setminus \mathcal{K}_i^\zeta|} (2^{|\mathcal{K}_i^\zeta|} - X)$.

Alternatively, we can also employ an SCL decoder with a sufficiently large decoding output list size L , which exhibits a computational complexity of $O(L \cdot N \log_2 N)$ for enumerating the MWCs [34]. But the SCL-based enumeration method is conducted when the all-zero codeword is transmitted at a very high SNR, e.g., 20 dB. Recalling Theorem 7, the list size should satisfy $L \geq 2^{|\mathcal{K}_i|}$ for enumerating all the MWCs in the coset $\mathcal{C}_i(\mathcal{I})$. Hence, computational complexity of this method is at least $O(2^{|\mathcal{K}_i|} \cdot N \log_2 N)$. Since $2^{|\mathcal{K}_i^\zeta|} \cdot (\zeta - i) \ll 2^{|\mathcal{K}_i|} \cdot N \log_2 N$, the proposed coset-wise enumeration method is much simpler than the existing SCL-based enumeration method.

V. AUB-OPTIMAL MPAC CODES

This section designs the AUB-optimal MPAC codes through searching for the optimal convolutional parameters (N_c^*, K_c^*) . This design approach shows a partially convolutional transformation can improve the minimum weight distribution for an MPAC code over its prototype PAC code, attributed to utilizing the row combinations of the frozen set more efficiently. The AUB-optimal MPAC code outperforms a PAC code and a CRC-polar code in terms of both MWD and decoding performance.

A. Design of AUB-Optimal MPAC Codes

With the proposed coset-wise analysis of the MWD for MPAC codes, the minimum distance $d_{\min}$ and the number of MWCs $A_{d_{\min}}$ can be determined. Furthermore, eq. (2) implies that both maximizing $d_{\min}$ and minimizing $A_{d_{\min}}$ can optimize the AUB P_e . The asymptotic ML decoding performance of different MPAC codes that are specified by their convolutional parameters

Algorithm 3: Enumeration of MWCs in coset $\mathcal{C}_i(\mathcal{I})$ of MPAC Code, i.e. $A_{i,\omega_{\min}}^{\text{MPAC}}$

Input: $i, \mathcal{I}_c, \mathcal{F}_c, \mathcal{I}_p, \mathcal{F}_p, S, \underline{g}_0^t$;
Output: $A_{i,\omega_{\min}}^{\text{MPAC}}$;

1 **Initialize** $X = 0$ and construct sets $\mathcal{K}_i$ and $\mathcal{F}^*$;
2 **If** $\mathcal{F}^* = \emptyset$ **then** // Remark 8
3 **Return** $A_{i,\omega_{\min}}^{\text{MPAC}} = 2^{|\mathcal{K}_i|}$;
4 **For every** $\mathcal{J} \subseteq \mathcal{K}_i$ **do**
5 Set $\tilde{\mathcal{J}}, \tilde{\mathcal{M}} \leftarrow \emptyset$;
6 Set $\underline{u}_0^{N-1} = \underline{0}_0^{N-1}, \underline{s}_0^{N_c-1} = \underline{0}_0^{N_c-1}$ and $S = \underline{0}_0^{t-1}$;
7 **If** $i \in \mathcal{I}_c$ **then**
8 Compute $k = F_{\mathcal{I}_c \cup \mathcal{F}_c}(i)$ and set $s_k = 1$;
9 $[u_i, S] \leftarrow \text{conv1b}(s_k, S, \underline{g}_0^t)$;
10 **For** $x = i + 1$ **to** $N - 1$ **do**
11 **If** $x \in \mathcal{I}_c$ or $x \in \mathcal{I}_p$ **then**
12 **If** $x \in \mathcal{J}$ **then** // for $x \in \mathcal{I}_c \cup \mathcal{I}_p$
13 $\tilde{\mathcal{M}} \leftarrow \text{addToM}(i, x, \tilde{\mathcal{J}}, \tilde{\mathcal{M}})$; // [41, Alg. 2]
14 $\tilde{\mathcal{J}} \leftarrow \tilde{\mathcal{J}} \cup \{x\}$;
15 **If** $x \in \mathcal{I}_c$ **then**
16 Compute $k = F_{\mathcal{I}_c \cup \mathcal{F}_c}(x)$;
17 **If** $x \in \mathcal{J}$ **then**
18 Set $u_x = 1$;
19 **Else**
20 **If** $x \in \tilde{\mathcal{M}}$ **then** set $u_x = 1$;
21 **If** $x \notin \tilde{\mathcal{M}}$ **then** set $u_x = 0$;
22 Compute $s_k = \lambda(u_x, S)$;
23 Update $S \leftarrow \text{conv1b}(s_k, S, \underline{g}_0^t)$;
24 **Else if** $x \in \mathcal{F}_c$ **then**
25 Compute $k = F_{\mathcal{I}_c \cup \mathcal{F}_c}(x)$;
26 Compute $[u_x, S] \leftarrow \text{conv1b}(0, S, \underline{g}_0^t)$;
27 **If** $x \in \mathcal{F}^*$ **then**
28 **If** $u_x = 1$ and $x \notin \tilde{\mathcal{M}}$ **then**
29 $X = X + 1$, **break**; // C1
30 **If** $u_x = 0$ and $x \in \tilde{\mathcal{M}}$ **then**
31 $X = X + 1$, **break**; // C2
32 **Else**
33 **If** $u_x = 1$ **then** // for $x \in \mathcal{F}_{\mathcal{J}}$
34 $\tilde{\mathcal{M}} \leftarrow \text{addToM}(i, x, \tilde{\mathcal{J}}, \tilde{\mathcal{M}})$;
35 $\tilde{\mathcal{J}} \leftarrow \tilde{\mathcal{J}} \cup \{x\}$;
36 **Else** // for $x \in \mathcal{F}_p$
37 **If** $x \in \tilde{\mathcal{M}}$ **then**
38 $X = X + 1$, **break**; // C3
39 **Return** $A_{i,\omega_{\min}}^{\text{MPAC}} = 2^{|\mathcal{K}_i|} - X$;

(N_c, K_c) can be evaluated by their corresponding AUBs P_e . It guides the design of MPAC codes, especially in determining (N_c, K_c) in optimizing the code's AUB performance. We call them the *AUB-optimal MPAC codes*.

Given N and K , let (N_c^*, K_c^*) denote the pair that optimizes the code's AUB performance. Considering the parameter constraint of (10), we have

$$\begin{aligned} 0 &\leq N_c - K_c \leq N - K, \\ 0 &< K_c \leq K, \\ 0 &< N_c \leq N. \end{aligned} \tag{31}$$

The above inequalities imply that $K_c \in [1, K]$ and $N_c \in [K_c, N - K + K_c]$. There are $K(N - K + 1)$ pairs of (N_c, K_c) . With the coset-wise analysis of MWD, this parameter search becomes practical. Since the two rate profilings and the choice of (N_c, K_c) jointly determine $d_{\min}$, maximizing $d_{\min}$ can be realized by choosing specific pairs of (N_c, K_c) . This helps rationalize the otherwise exhaustive search process.

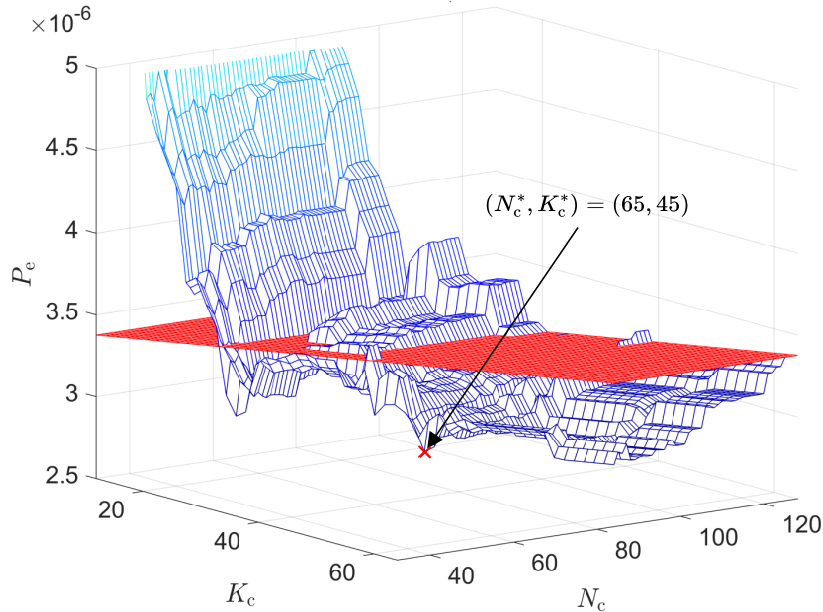


Fig. 8. AUB performances of the MPAC codes and PAC codes versus parameters (N_c, K_c) , where $N = 128$, $K = 64$ and $E_b/N_0 = 3.5$ dB.

Fig. 8 shows how the AUB performances P_e of MPAC codes are affected by K_c and N_c under the SNR per information bit of $E_b/N_0 = 3.5$ dB. We consider the MPAC codes with length $N = 128$, dimension $K = 64$ and the convolutional generator polynomial coefficients of $(1, 0, 1, 1, 0, 1, 1)$. The rate profiling is conducted by GA at an SNR of 0 dB. We compare the

AUB performances of MPAC codes with the prototype (128, 64) PAC code of [7]. The PAC code is constructed by the RM rate profiling. Its MWD is $d_{\min} = 16$ and $A_{d_{\min}} = 3120$. The AUB performances are measured versus parameters (N_c, K_c) , yielding the blue and the red meshes that correspond to the MPAC codes and the PAC code, respectively. In particular, the AUB-optimal MPAC code with parameter $(N_c^*, K_c^*) = (65, 45)$ is marked by a red ‘ $\times$ ’ symbol. The code has $d_{\min} = 16$ and $A_{d_{\min}} = 2633$. It can be seen that there exist numerous (N_c, K_c) that can enable an MPAC code to outperform the prototype PAC code. This is due to a better weight distribution is produced by the partially convolutional transform in the MPAC codes. Furthermore, Fig. 8 also suggests that a larger K_c is more likely to yield a better weight distribution for the MPAC codes. It helps maximize $d_{\min}$ and yields a greater MWC number deficit X . With a large K_c , reducing N_c within the scope that keeps $d_{\min}$ being maximized generally achieves a better AUB performance. This indicates that assigning elements from set $\mathcal{F}_c$ to set $\mathcal{F}_p$ can lead to a better MWD for the MPAC codes. An extreme case to the opposite is when $K_c = 0$, the MPAC codes become polar codes with no improvement on its weight distribution.

B. Simulation Results

This subsection presents the MWD results and performances of the AUB-optimal MPAC codes. The AUB-optimal MPAC codes are also compared with the relevant coding schemes, especially the MPAC codes designed in Section III. Note that the parameters of the relevant coding schemes that are applied in this subsection are the same as those in Section III-C.

TABLE I
THE MWD OF AUB-OPTIMAL MPAC CODES, PAC CODES, ENSEMBLE AVERAGE AND CRC-POLAR CODES WITH DIFFERENT CODE PARAMETERS

N	K	PAC		MPAC			Ensemble Average		CRC-polar	
		$d_{\min}$	$A_{d_{\min}}$	(N_c^*, K_c^*)	$d_{\min}$	$A_{d_{\min}}$	$d_{\min}$	$A_{d_{\min}}$	$d_{\min}$	$A_{d_{\min}}$
128	29	32	820	(61,29)	32	436	32	422.1	24	53
	64	16	3120	(65,45)	16	2633	16	2766.9	12	224
	99	8	16080	(89,71)	8	15264	8	15936.0	6	152
256	93	32	5092	(145,86)	32	2788	32	2766.9	16	6
	163	16	18896	(169,132)	16	15622	16	15936.3	8	4
512	130	64	17092	(252,110)	64	3466	64	2766.9	32	256
	256	32	35900	(342,222)	32	17108	32	15936.3	16	966
	382	16	128704	(314,216)	16	104209	16	101280.0	8	2307

Table I provides the MWD of PAC codes, the AUB-optimal MPAC codes, the ensemble average of pre-transformed polar codes [30] and CRC-polar codes. The average MWD of pre-transformed polar codes is analyzed by the probabilistic method that calculates the weight spectrum of the pre-transformed polar codes with randomly generated pre-transformed matrix and the RM rate profiling. The detailed procedure of this method can be found in [30]. MPAC codes are constructed by the proposed iRMP construction, while PAC codes are constructed by the RM rate profiling. The convolutional generator polynomial coefficients $(1, 0, 1, 1, 0, 1, 1)$ are employed. The CRC-polar codes are constructed by the 5G reliability sequence.

It can be seen that for a wide range of code parameters, the AUB-optimal MPAC codes yield a better MWD than the respective PAC codes and CRC-polar codes. In particular, the AUB-optimal MPAC codes yield the same minimum distance $d_{\min}$ as the PAC codes, but they contain less MWCs. Meanwhile, it can also be observed that the minimum distance of the AUB-optimal MPAC codes are greater than that of the CRC-polar codes. Furthermore, the AUB-optimal MPAC codes yield a close $A_{d_{\min}}$ compared to the average pre-transformed polar codes.

They demonstrate that the AUB-optimal MPAC codes have the MWD advantage over both the PAC codes and the CRC-polar codes.

With code parameters of N and K concerned in Table I, MPAC codes have the same information set $\mathcal{I}$ as the corresponding PAC codes. Their main difference is reflected in pre-transformed design. By defining $\underline{d}$ as bits before polar transformation, the pre-transformed matrix $\mathbf{G}_{\text{pre}}$ can be utilized to represent the constraint relationship between bits in $\underline{d}_0^{N-1}$. Note that $\underline{c}_0^{N-1} = \underline{d}_0^{N-1} \mathbf{G}_{\text{pre}} \mathbf{G}_p$. In particular, for PAC codes, we have $\mathbf{G}_{\text{pre}} = \mathbf{G}_c$. For MPAC codes, the submatrix of $\mathbf{G}_{\text{pre}}$ with rows and columns selected in $\mathcal{P}$ is $\mathbf{G}'_c$. $\mathbf{G}_{\text{pre}}$ is further obtained with diagonal elements set to 1 and other elements set to 0.

Therefore, the $\mathbf{G}_{\text{pre}}$ of an MPAC code is sparser than that of a PAC code and generally not a Toeplitz matrix. Consequently, some bits of an MPAC code in $\underline{d}_{\mathcal{I}_c}$ can influence other bits with a more distant index in $\underline{d}_{\mathcal{F}_c}$ than the case in PAC code. For low rate polar-like codes, the continuous length of the frozen set is large. With short convolutional generator polynomials employed, PAC codes can only utilize a portion of $\underline{d}_{\mathcal{F}_c}$ to reduce the number of MWCs sufficiently. However, the MPAC codes can utilize the sparser pre-transformed matrix to involve more rows in $\mathcal{F}_c$ in row combinations. This results in a better effect on the MWC number deficit X increasing especially for low code rates. As a result, the MPAC codes not only inherit the advantage of PAC codes, but also incorporate the row combinations of the frozen set. After selecting the convolutional

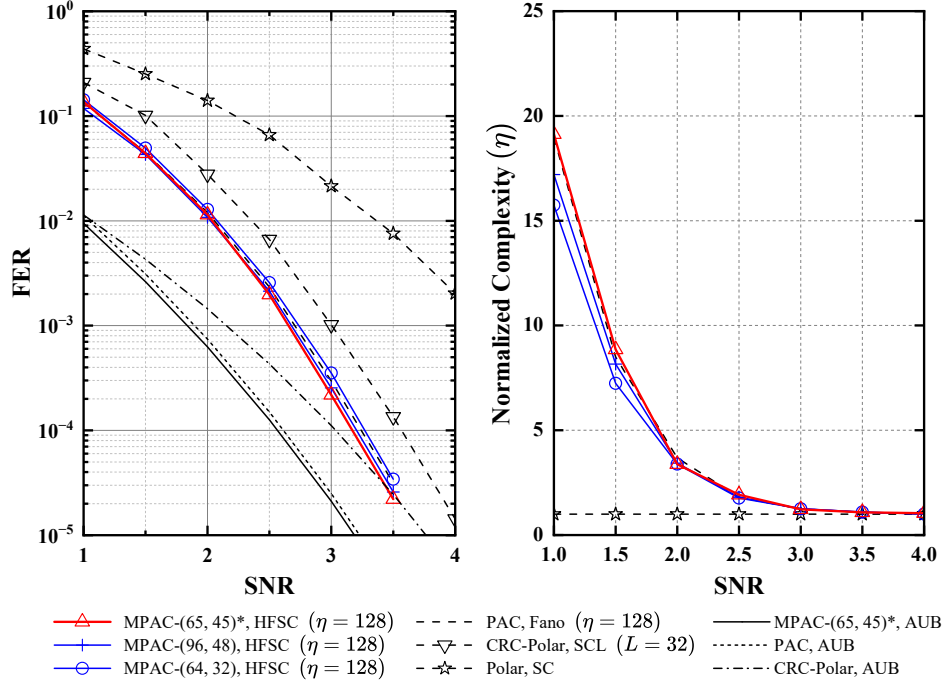


Fig. 9. Performance comparison of the AUB-optimal MPAC codes and other relevant codes, where $N = 128$ and $K = 64$.

parameters, this leads to a better MWD for the AUB-optimal MPAC codes than the PAC codes and the CRC-polar codes.

Note that as the code rate increases, the MWD improvement margin of MPAC codes becomes less significant. This is because in such situation, some MWCs in the specific cosets cannot be reduced.

The proportion of these irreducible MWCs increases with the code rate.

On the other hand, higher rate restricts the size of $\mathcal{F}_p$ for MPAC codes, which leads to less improvement of MWD in comparison with PAC codes.

Finally, Fig. 9 shows the HFSC decoding FER, normalized complexity and AUB of the optimized MPAC code with $N = 128$ and $K = 64$. A decoding computation threshold of $\eta = 128$ is applied. The decoding FER and complexity performances of different coding schemes are also provided, including the MPAC-(96, 48) and the MPAC-(64, 32) codes. Note that the MWD of the MPAC-(96, 48) code and the MPAC-(64, 32) code are $\{d_{\min} = 16, A_{d_{\min}} = 2900\}$ and $\{d_{\min} = 16, A_{d_{\min}} = 3121\}$, respectively. It can be seen that the AUB-optimal MPAC code indicated by * as MPAC-(65, 45)* can slightly outperform the MPAC-(96, 48) code and the

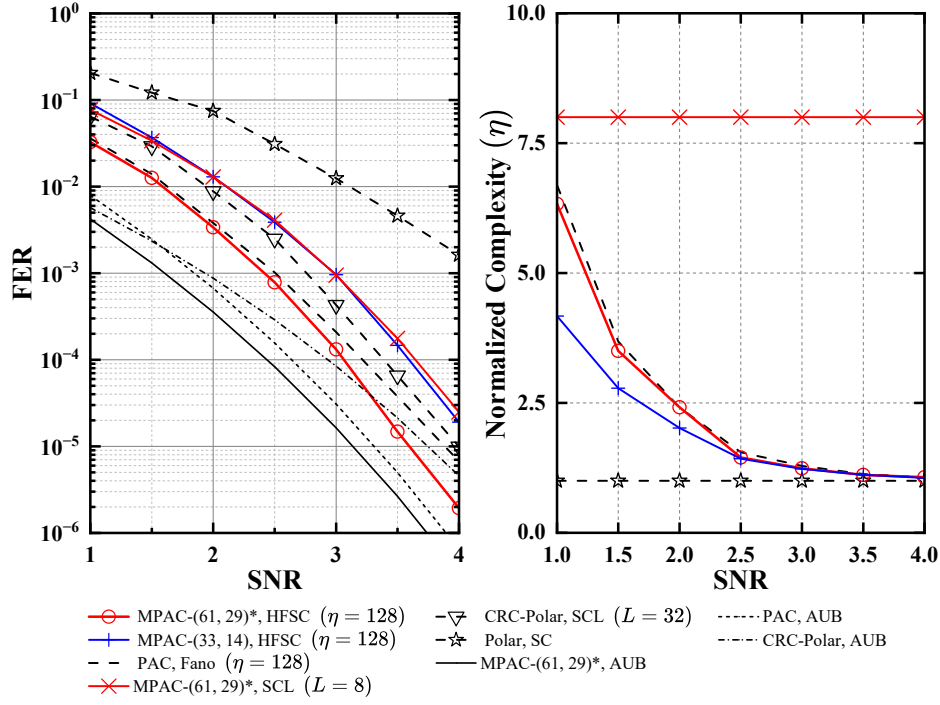


Fig. 10. Performance comparison of the AUB-optimal MPAC codes and other relevant codes, where $N = 128$ and $K = 29$.

MPAC-(64, 32) code with an increased complexity. Fig. 10 shows the same coding schemes with $N = 128$ and $K = 29$. It can be seen that the AUB-optimal MPAC code with HFSC decoding outperforms the MPAC-(33, 14) code, the PAC code and the CRC-polar code in both the decoding FER and the AUB.

VI. CONCLUSION

This paper has proposed the MPAC codes and their HFSC decoding.

Realized by two rate profiling steps with convolutional parameters, a subset of the information bits first undergo the convolutional transform. The convolutional output then together with the remaining information bits undergo the polar transform. Consequently, the HFSC decoding deploys the Fano decoding and SC decoding to recover the information bits that have undergone the convolutional transform and those that have not, respectively. The MWD analysis of the MPAC codes has been provided in the coset-wise framework, which can be further utilized to optimize the design of MPAC codes, especially in selecting the convolutional parameters.

Consequently, the AUB-optimal MPAC codes can be designed, achieving a better MWD than the PAC codes and CRC-polar codes. The implication of the MWD improvement is examined

by effectively exploiting the row combinations of the frozen set, which originates from the sparser pre-transformed matrix of MPAC codes. Simulation results of the proposed schemes and the relevant schemes have been presented to verify the decoding performance and complexity advantages of the proposed MPAC coding schemes.

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